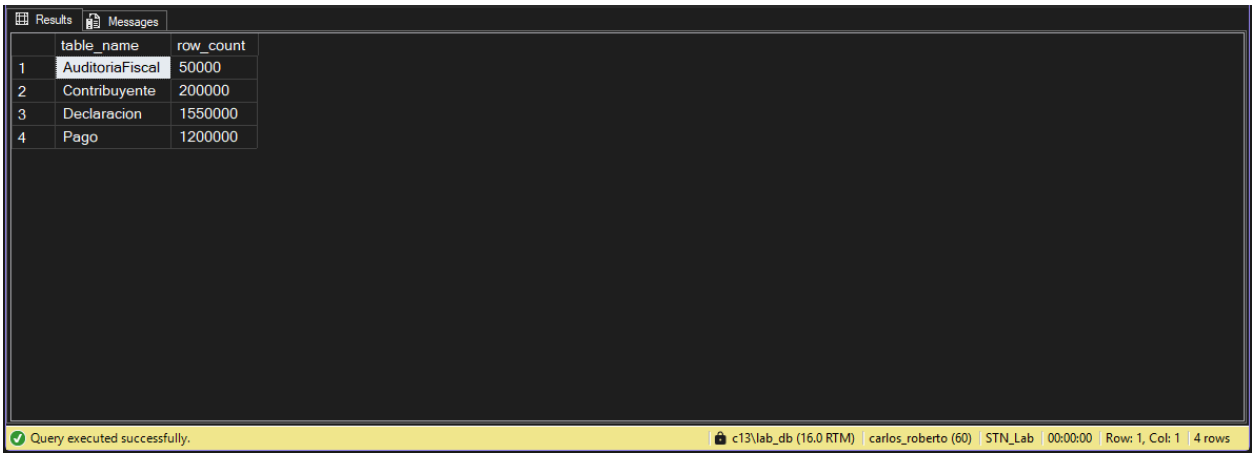
01_clustered_vs_nonclustered

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

1.0 ENVIRONMENT CHECK

```
USE STN_Lab;
GO
-- ============================================================
==
-- Verify lab tables exist before running any section.
-- ============================================================
==
SELECT
    t.name        AS table_name,
    p.rows        AS row_count
FROM sys.tables t
JOIN sys.partitions p
    ON t.object_id = p.object_id
   AND p.index_id IN (0, 1)
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
ORDER BY t.name;
GO
```



Results | Messages

	table_name	row_count
1	AuditoriaFiscal	50000
2	Contribuyente	200000
3	Declaracion	1550000
4	Pago	1200000

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (60) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 4 rows

1.1 ESTADO ACTUAL DE ÍNDICES

```
-- ============================================================
==
-- 1.1 ESTADO ACTUAL DE ÍNDICES
-- Fotografía del estado antes de crear nada.
-- Ejecutar antes y después de cada CREATE INDEX.
-- ============================================================
==
SELECT
    t.name                          AS table_name,
    i.name                          AS index_name,
    i.type_desc                     AS index_type,
    i.is_unique,
    i.is_primary_key,
    STRING_AGG(c.name, ', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal)
                                    AS key_columns
FROM sys.indexes i
JOIN sys.tables t
    ON i.object_id = t.object_id
JOIN sys.index_columns ic
    ON i.object_id = ic.object_id
   AND i.index_id  = ic.index_id
JOIN sys.columns c
```

```
    ON ic.object_id = c.object_id
   AND ic.column_id = c.column_id
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago')
  AND ic.is_included_column = 0
GROUP BY t.name, i.name, i.type_desc, i.is_unique, i.is_prima
ry_key
ORDER BY t.name, i.type_desc DESC;
GO
```

Results | Messages

	table_name	index_name	index_type	is_unique	is_primary_key	key_columns
1	Contribuyente	IX_Contribuyente_NIT	NONCLUSTERED	0	0	NIT
2	Contribuyente	IX_Contribuyente_NIT_Covering	NONCLUSTERED	0	0	NIT
3	Contribuyente	PK_Contribuyente	CLUSTERED	1	1	ContribuyenteID
4	Declaracion	IX_Declaracion_ContribuyenteID	NONCLUSTERED	0	0	ContribuyenteID
5	Declaracion	PK_Declaracion	NONCLUSTERED	1	1	DeclaracionID
6	Pago	PK_Pago	NONCLUSTERED	1	1	PagoID

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (60) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 6 rows

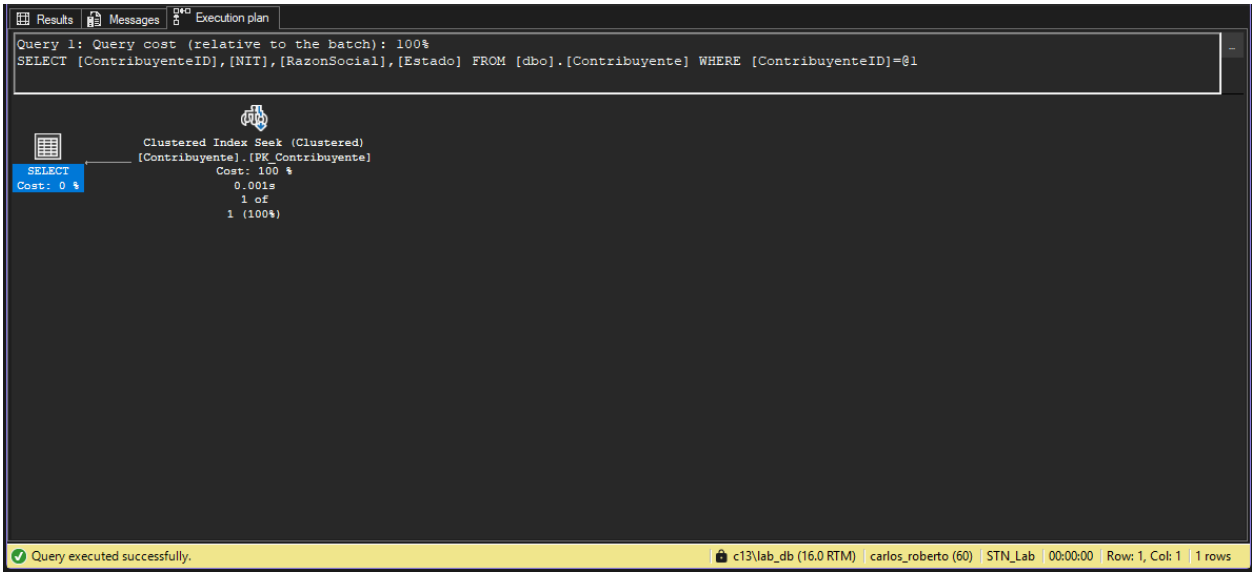
1.2 CLUSTERED - NONCLUSTERED

INDEX SEEK POR CLAVE REAL (ContribuyenteID)

```
SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT
    ContribuyenteID,
    NIT,
    RazonSocial,
    Estado
FROM dbo.Contribuyente
WHERE ContribuyenteID = 500;   -- reemplazar con ID existente
en el lab
```

```
SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



- ContribuyenteId es la clave Clustered.
- Buscar por esta columna garantiza Clustered Index Seek puro: el motor navega el B-tree y llega directo al dato.
- Sin Key Lookup posible el nivel hoja tiene toda la fila.

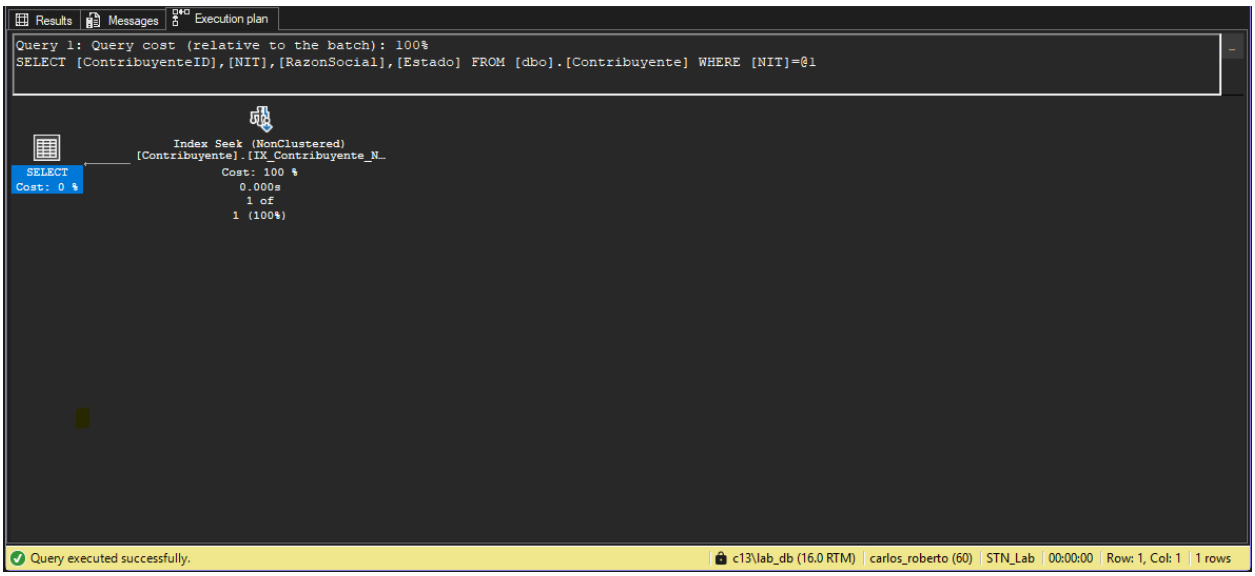
NON-CLUSTERED SEEK POR NIT

```
SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT
    ContribuyenteID,
    NIT,
    RazonSocial,
    Estado
FROM dbo.Contribuyente
```

```
WHERE NIT = '2010219';   -- reemplazar con NIT existente

SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



NIT tiene su propio NC index. Al buscar por NIT el optimizador usa ese índice — correcto.

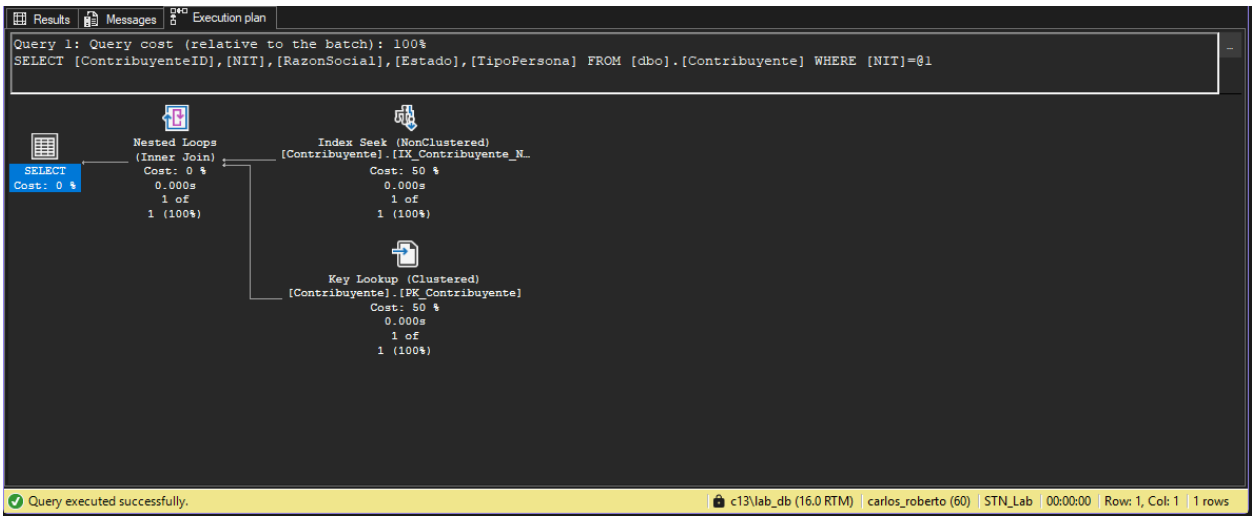
NC COVERING SEEK POR NIT

```
SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT
    ContribuyenteID,
    NIT,
    RazonSocial,
    Estado,
    TipoPersona  --se agrega esta columna que no esta cubiert
a en ningun indice
FROM dbo.Contribuyente
```

```
WHERE NIT = '2010219';   -- reemplazar con NIT existente

SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```

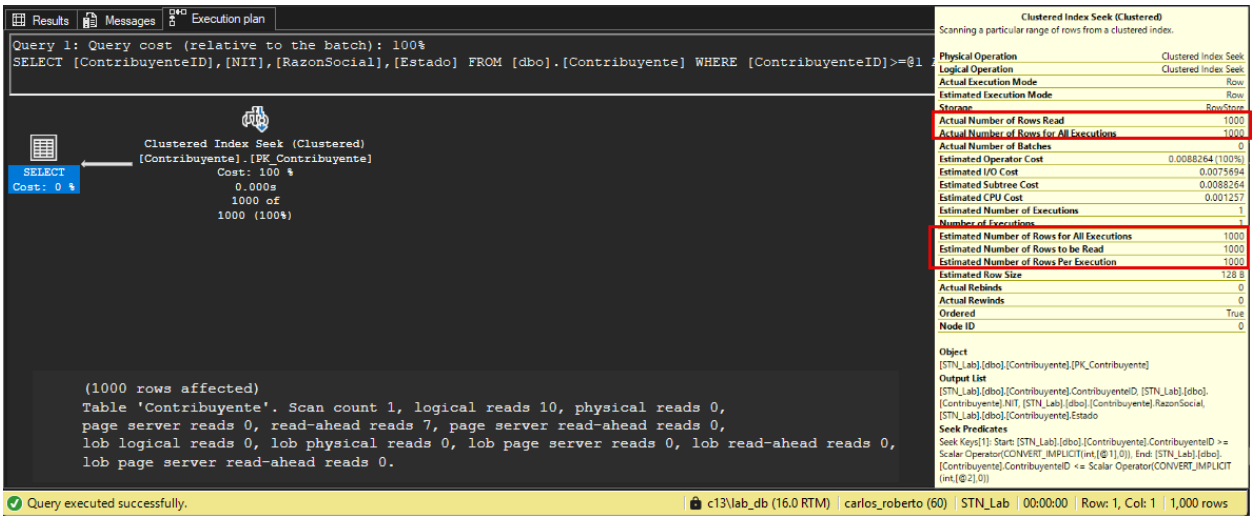


1.3 CLUSTERED INDEX RANGE SCAN

```
SET STATISTICS IO ON;

SELECT
    ContribuyenteID,
    NIT,
    RazonSocial,
    Estado
FROM dbo.Contribuyente
WHERE ContribuyenteID BETWEEN 1 AND 1000;

SET STATISTICS IO OFF;
GO
```

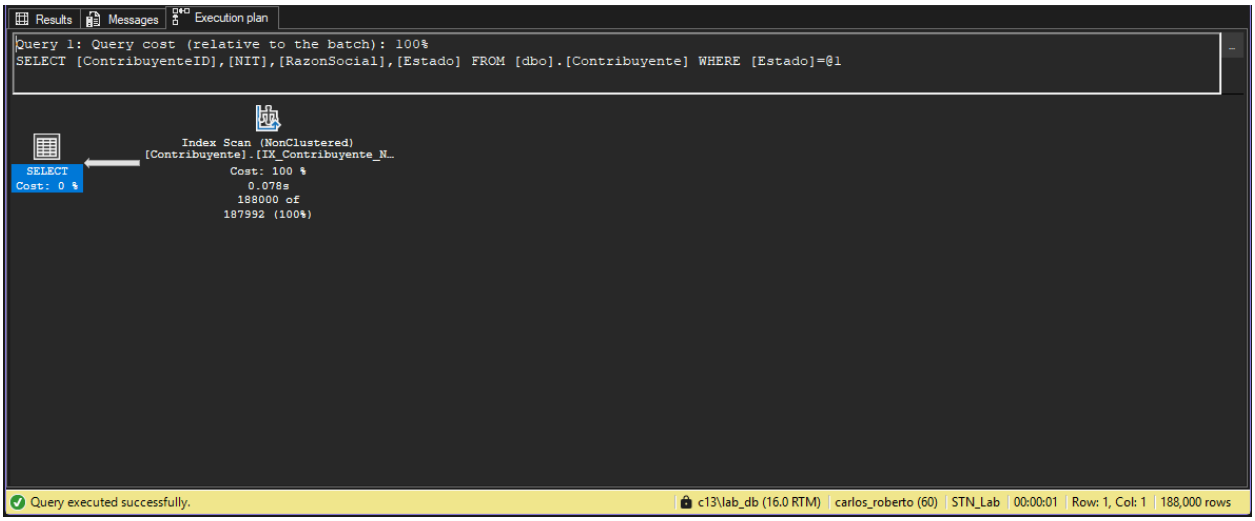


1.4 NON-CLUSTERED SIN ÍNDICE (Estado)

```
SET STATISTICS IO ON;

SELECT
    ContribuyenteID,
    NIT,
    RazonSocial,
    Estado
FROM dbo.Contribuyente
WHERE Estado = 'A';

SET STATISTICS IO OFF;
GO
```



Es el costo real de no tener índice en Estado.

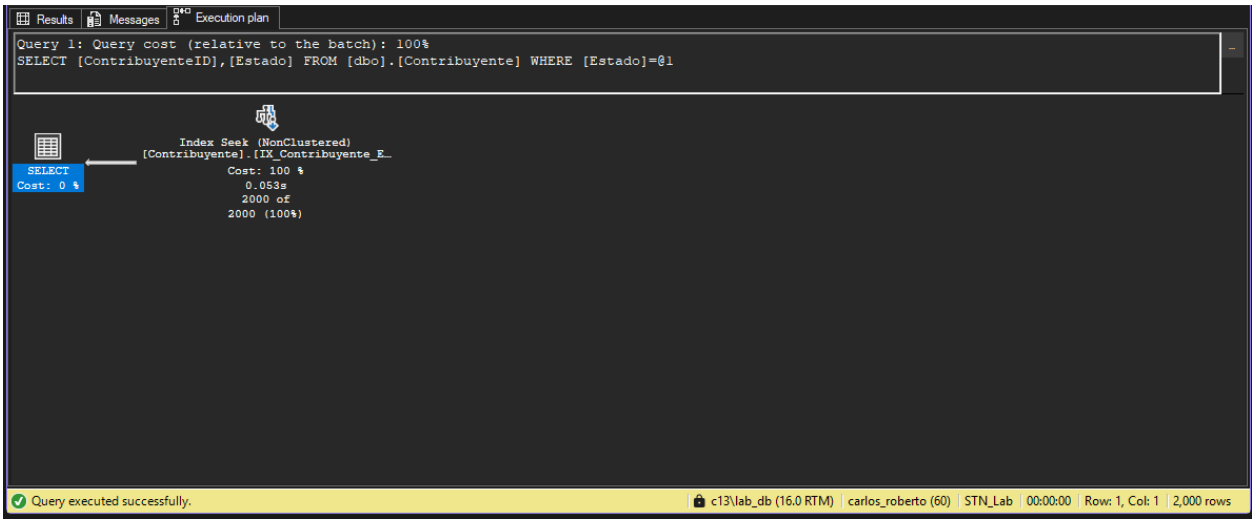
1.5 CREAR NC INDEX EN Estado

```
CREATE NONCLUSTERED INDEX IX_Contribuyente_Estado
    ON dbo.Contribuyente (Estado);
GO
```

```
SET STATISTICS IO ON;

SELECT
    ContribuyenteID,
    Estado
FROM dbo.Contribuyente
WHERE Estado = 'S';

SET STATISTICS IO OFF;
GO
```



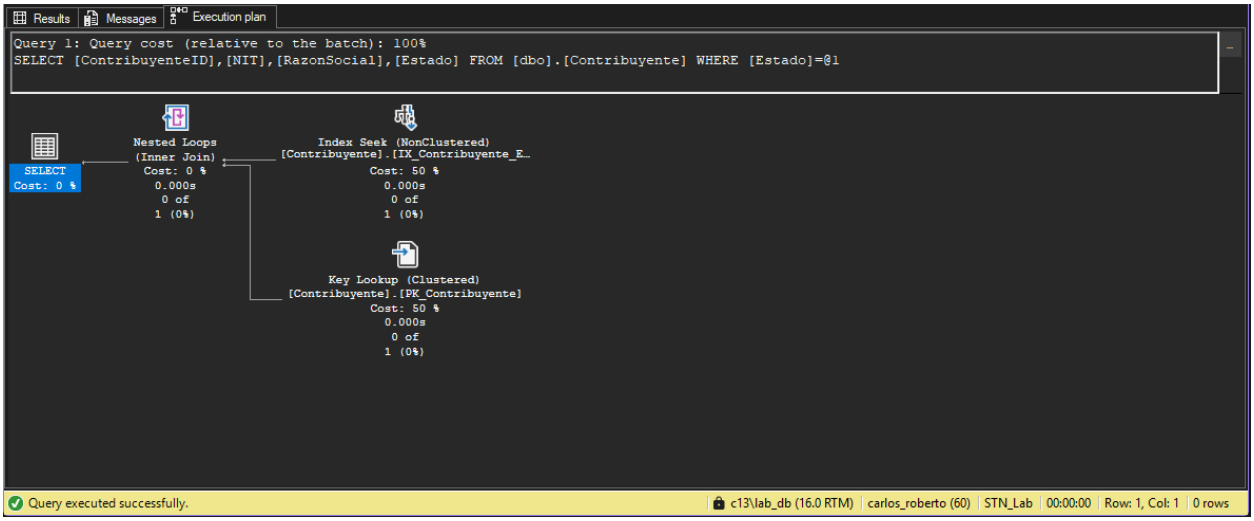
1.6 NC SEEK + KEY LOOKUP (Estado con índice)

Mismo query que 1.4. El optimizador ahora puede usar el NC. Pero RazonSocial y NIT no están en el índice = Key Lookup.

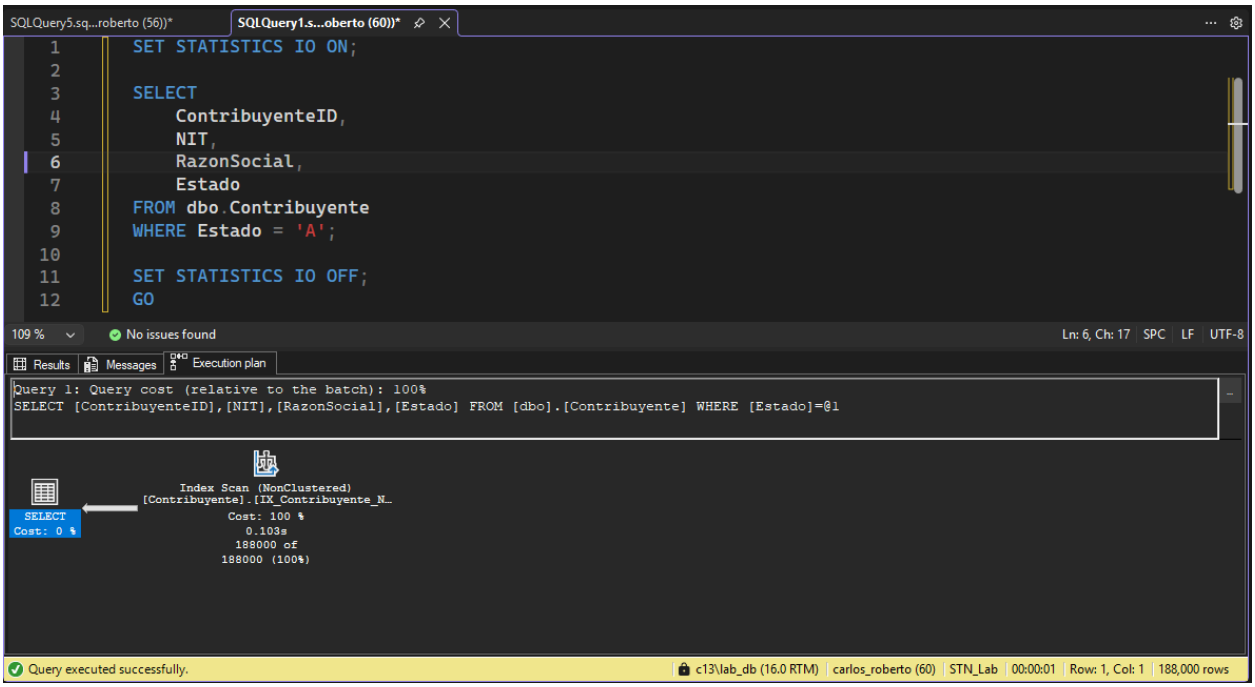
```
SET STATISTICS IO ON;

SELECT
    ContribuyenteID,
    NIT,
    RazonSocial,
    Estado
FROM dbo.Contribuyente
WHERE Estado = 'B';

SET STATISTICS IO OFF;
GO
```



Si el 80% de estado son A, el optimizador puede ignorar el NC y hacer Scan de todas formas.



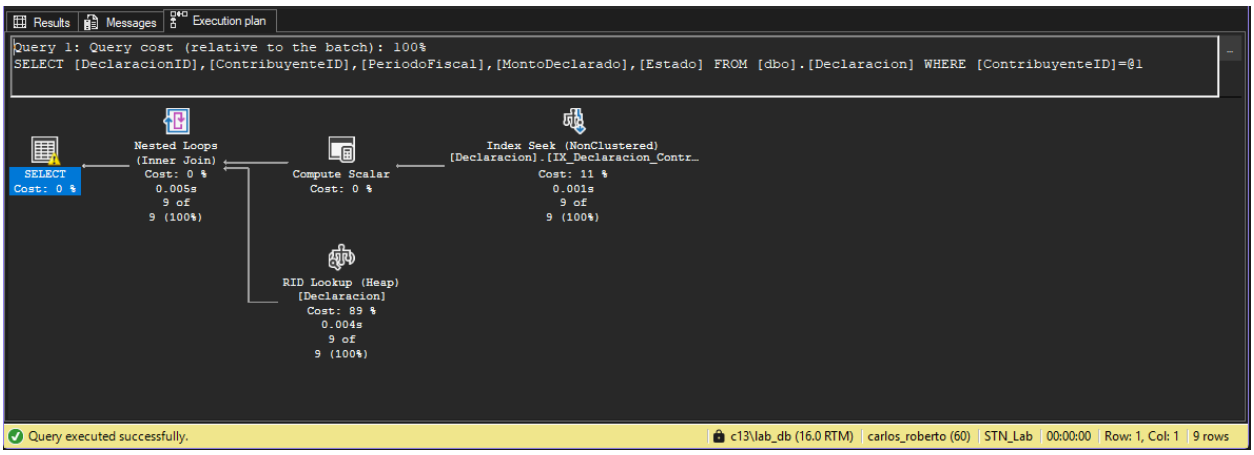
Un índice Non-Clustered solo produce Index Seek cuando el optimizador puede resolver la query sin salir del índice. En el momento en que la query pide columnas que no están en el índice, el motor calcula el costo de los Key Lookups por cada fila encontrada y lo compara contra leer toda la tabla. Si los Lookups cuestan más, elige el Scan y tiene razón. La solución no es forzar el Seek, es eliminar la necesidad del Lookup con columnas INCLUDE.

1.7 — DECLARACIONES

```
SET STATISTICS IO ON;

SELECT
    DeclaracionID,
    ContribuyenteID,
    PeriodoFiscal,
    MontoDeclarado,
    Estado
FROM dbo.Declaracion
WHERE ContribuyenteID = 500;

SET STATISTICS IO OFF;
GO
```



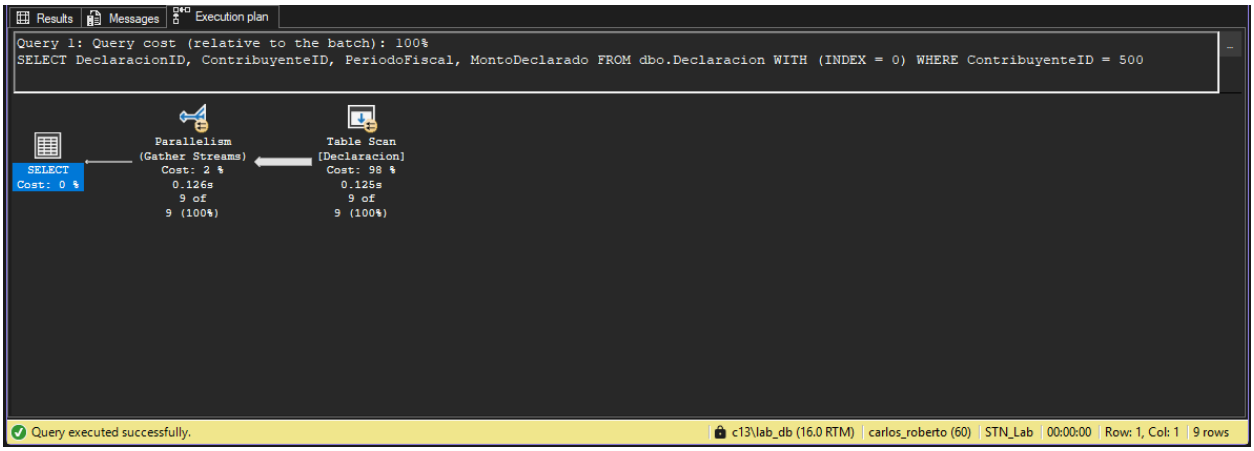
OBSERVACIÓN: el Seek encontró rápido, pero pagó Key Lookup por cada columna fuera del índice.

1.8 FORZAR OPERADORES (solo laboratorio)

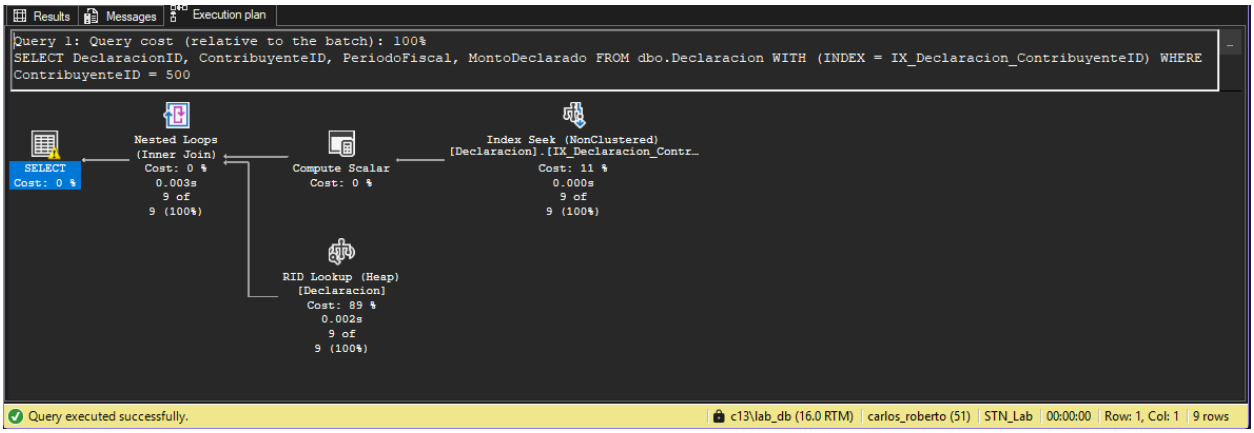
```
-- Forzar lectura directa del heap (sin usar ningún NC index)
SET STATISTICS IO ON;
```

```
SELECT
    DeclaracionID,
    ContribuyenteID,
    PeriodoFiscal,
    MontoDeclarado
FROM dbo.Declaracion WITH (INDEX = 0)
WHERE ContribuyenteID = 500;

SET STATISTICS IO OFF;
GO
```




```
-- Forzar NC index — verá RID Lookup porque la tabla es heap
SET STATISTICS IO ON;

SELECT
    DeclaracionID,
    ContribuyenteID,
    PeriodoFiscal,
    MontoDeclarado
FROM dbo.Declaracion WITH (INDEX = IX_Declaracion_Contribuyen
teID)
WHERE ContribuyenteID = 500;
```

```
SET STATISTICS IO OFF;
GO
```